L.J. INSTITUTES OF COMPUTER APPLICATIONS

Programme	Master of Computer Applications					**Branch/Spec.**	Computer Applications		
Semester	II					**Effective from Academic Year**	2025-26		
Subject Category	Major					**Effective for the batch Admitted in**	June 2025		
Subject Code	240110204		**Subject Name**			Python Programming (Python)			
Teaching scheme						**Examination scheme (Marks)**			
(Lab hours Per week)	**Credit**	**L**	**T**	**P**	**Total**	**E**	**CEC**	**V**	**Total**
6	3	0	0	6	72	0	50	50	100

University Exam Evaluation Scheme

Internal Evaluation		**External Evaluation**	
Mid Sem (50 Marks)	**30**	**Theory Exam**	**NA**
Attendance	**10**	**Practical Exam**	**50**
Hacker Rank Practical Assignment	**10**		
Total Marks	**50**	**Total Marks**	**50**

Pre-requisites:

Knowledge of the C programming language.

Program Learning Outcome:

Name of PO	**Description**
PO1	Foundational & Theoretical Mastery
PO2	Software Design & Development Expertise
PO3	Problem-Solving & Analytical Ability
PO4	Proficiency in Emerging Technologies
PO5	Information Systems & Management Perspective
PO6	Professional & Ethical Responsibility
PO7	Communication & Teamwork
PO8	Lifelong Learning & Career Readiness

Course Learning Outcome:

Name of CO	**Description**
CO1	To understand the basics of Python, including its history, applications, advantages, and disadvantages. Set up the Python environment and work with both interactive and script modes, including Python tokens, keywords, and identifiers.
CO2	To implement conditional statements, loops, and functions in Python. Learn about recursion and exception handling techniques, including the use of try, except, raise, and finally blocks to ensure robust code.

CO3	To understand and work with Python's core data structures (lists, tuples, dictionaries, and strings). Perform file operations such as reading from and writing to text files, and handle CSV and Excel files using Python
CO4	To use Pandas to manipulate and analyze data, including handling missing data, grouping, merging, and aggregating. Create visualizations using Matplotlib to represent data through various plots (scatter, bar charts, histograms, etc.).
CO5	To integrate Python concepts learned (file handling, data analysis, visualization, and database connectivity) to develop real-world data-driven applications. This includes using MySQL for database interaction and creating practical solutions to problems.

Mapping of CO and PO:

Course Learning Objectives	Program Outcomes (POs)							
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	0	3	0	0	2	2
CO2	3	3	0	3	0	0	0	0
CO3	3	2	0	3	0	0	0	0
CO4	0	2	0	3	2	0	0	0
CO5	2	3	0	3	0	0	0	3

Content:

Unit	Content	Weightage (%)	No of Hrs.	CO
Unit I	**Introduction to Python and Core Basics** • History and need of Python • Applications of Python • Advantages and Disadvantages of Python • Installing Python • Program Structure • Interactive Shell • Executable / Script files • User Interface / IDE (IDLE, SPYDER) • Working with Interactive Mode • Working with Script Mode **Python Fundamentals** • Python Character Set • Python • Tokens • Keywords and Identifiers • Variables , Constants and Assignments **Basic Data Types** • Numbers Representing binary, octal and hexadecimal numbers in python • Strings • Basic operations on data types	20%	14	CO1

	• Introduction to Mutable and Immutable Data Types Basic Input and Output in Python **Operators and Expressions** Basic Operators –Arithmetic operators, Relational operators, Logical operators, Membership operators, Ternary operators, Boolean operators, Bitwise Operators. • Type Conversion • Expressions and Evaluation • Formatted Input and Output			
Unit II	**Program Control Flow and Python Data Structures** **Conditional Statements** o if o if-else o if-elif o Nested if • Python Indentation **Looping and Iteration** • range() function o Introduction o Types o Use of range() • for loop • while loop • loop-else statement • Nested loops • Break and Continue **Immutable Data Structures** – Strings, Tuples **Mutable Data structures** – Sets, Lists and Dictionaries	**20%**	**14**	**CO2**
Unit III	**Functions and Files in Python** **• Functions** • Built-in Functions • Introduction to Functions • Structure of Python Functions • User Defined Functions (UDF) • Arguments and Parameters o Default Arguments o Named Arguments • Scope of Variables • Lambda Functions • Recursion Functions **File Operations** • Text and Byte Files • Opening Files • Reading and Writing Files • Other File Tools	**20%**	**14**	**CO3**

	• Working with MS Excel Files
	    ○ `.xls`
	    ○ `.csv`			

Unit IV	**Exception Handling and Database Connectivity and GUI** **Exception Handling** • Default Exceptions and Errors • try–except statement • raise statement • assert • finally block • User Defined Exceptions **Database Connectivity** • Introduction to MySQL • PyMySQL Connections • Executing Queries • Transactions • Handling Database Errors	**20%**	**14**	**CO4**

UNIT V	**Data Science using Python- Pandas and Matplotlib**
	Pandas
	• Importing data from
	    ○ CSV
	    ○ Excel
	    ○ HDF5
	    ○ SQL Databases
	• Series and DataFrames
	• Multilevel Series
	• Grouping and Aggregation
	• Generating Summary Tables
	• Data Wrangling
	• Merging and Joining Data Frames
	• Date and Time Manipulation
	• Creating Metrics for Analysis
	Data Visualization using Matplotlib
	• Scatter Plot
	• Bar Chart
	• Histogram
	• Stack Charts
	• Legends, Titles, Styles
	• Figures and Subplots
	• Plotting with Pandas
	• Labelling and Arranging Figures
	• Saving Plots
		20%	**14**	**CO5**

Reference Books:	
1	Core python programming by Dr. R. Nageshwar Rao
2	Pandas for Everyone by Daniel Y. Chen
3	McKinney Wes, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython", O'Reilly Media, 2012.
4	Hauck Trent, "Instant Data Intensive Apps with Pandas How-To", Packt Publishing Ltd, 2013.
Web Reference:	
1	https://www.python.org
2	https://www.w3schools.com/python
Question Paper Scheme:	
	University Examination Duration: 2 Hours 30 minutes. **SECTION-I** PRACTICAL – (40 marks) **SECTION-II** VIVA (10 Marks)